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Pharmaceutical wastewater toxicity: An ignored threat to the public health

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ABSTRACT

Pharmaceutical waste disposal is a significant problem for the global healthcare system. It poses a threat to water safety, which is already a challenge as 1 out of 3 people still lack access to safe drinking water. We conducted a thorough literature review of relevant articles published between 2000 and 2023. Our review scrutinized 107 studies, each of which applied preset inclusion and exclusion criteria to ensure highquality research. The focus of our interest was on the health risks posed by pharmaceutical compounds in waste. We also evaluated recycling methods and proposed practical measures for proper education and the implementation of strict legislation to address this issue. Pharmaceutical compounds have been found in water resources worldwide. These compounds can have negative effects on multiple organs and it's important to remove them from water resources. However, water quality varies globally due to limitations in accessibility and budget for water purification and recycling efforts. Comprehensive studies are needed to identify drinking water's most common pharmaceutical residues and develop effective removal strategies. Proper disposal of pharmaceuticals is crucial to protect water resources and ensure clean drinking water. This requires raising awareness, educating the public population, implementing strict regulations, and collaborating among stakeholders.

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Disposal; ecotoxicity; health; pharmaceutical; regulation; recycling; safety; toxicity; wastewater; drugs

1. Introduction

Pharmaceuticals are a significant class of chemical compounds used to diagnose, prevent, and treat ailments in humans and animals (Haider, 2023; Paszkiewicz et al., 2022). The global population growth, economic improvement, the rise of animal husbandry, the increasing number of newlydiscovered drugs, and the development of healthcare systems have led to a tremendous increase in the production and use of pharmaceuticals, resulting in more significant amounts of pharmaceutical waste disposal (Awathale & Kokare, 2023; Bharti & Bora, 2023; Cobongela, 2023; Costa et al., 2019; Desai et al., 2022; Mahmood Khan et al., 2023). According to the World Health Organization (WHO), pharmaceutical waste residues refer to “all expired pharmaceuticals, unsealed syrups or eye drops, or tubes of cream, ointment, all bulk loose tablets and capsules, as well as cold chain, damaged

unexpired pharmaceuticals that should have been stored in a cold chain but were not” (Ariffin & Zakili, 2019).

Pharmaceutical residue was first reported in watersheds in 1976 in Kansas City, US, with Clofibric acids being found in concentrations ranging from 0.8 to 2.2 µg/L. In 1981, pharmaceutical residue was also discovered in river waters in the United Kingdom (UK), and in 1986, high concentrations of ibuprofen and naproxen were detected in Canada (Ebele et al., 2017; Metcalfe et al., 2004). Since then, several sources that could potentially release pharmaceutical waste into the water have been discussed. One of the most significant sources of residue release is excretions, such as feces and/or urine, as human and animal bodies rarely metabolize and ingest medications (Rogowska & Zimmermann, 2022). Other direct sources include municipal, agricultural, domestic, health-care, and research activities, as well as animal waste, coastal waters, and industries (Anwar et al., 2021;

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Morosini et al., 2020; Фрумин & Frumin, 2022). The primary ecological problem we are currently facing is water shortage brought on by rising water pollution. According to reports, 71% (4.3 billion) of the world's population experiences water scarcity for a few months each year (Jaiswar et al., 2022).

The increased use and release of pharmaceuticals into watersheds have become a global concern due to the potential adverse effects on humans, animals, and the environment. Pharmaceuticals are not easily biodegradable in environment and lot of factors effect on their degradation (Guo et al., 2017). These properties make these chemicals toxic, even at low-to-moderate concentrations, and can cause complications by increasing antibiotic resistance (Latif et al., 2023), disturbing the endocrine (Kairigo et al., 2020) and immune systems (El Marghani et al., 2014), and affecting different body organs. The persistent presence of pharmaceutical waste in water bodies, soil, and other ecosystems can also threaten aquatic life by deteriorating water quality and reducing available oxygen (Brausch et al., 2012; Kock et al., 2023). These issues are more pronounced in regions with water scarcity where water is frequently reused (El Marghani et al., 2014).

Water purification/recycling may not always be feasible due to limited accessibility and budget. This article reviews studies reporting pharmaceutical residue concentration in water bodies and their toxic effects on nature, specifically humans. It also considers the pros and cons of available recycling methods, existing regulations, and the role of education. Lastly, it introduces feasible approaches for removing these chemicals' residue to combat this public health challenge.

2. Review methodology

A systematic literature review was conducted for the present study, covering the published articles from 2000 to 2023 in compliance with the guidelines of Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). The focus was on collecting and examining various literature on the subject of toxicity, as well as exploring the potential impacts of pharmaceutical wastewater on public health. As shown in Figure 1, the review process involved full-text reviewing of 107 articles that met the inclusion criteria. We conducted a comprehensive search using scientific databases such as PubMed/MedLine, Web of Science, Scopus, and other relevant databases. The keywords used for the literature search included: "Environmental health", "Drug-contaminated wastewater", "Medicinal wastewater", "Pharmaceutical wastewater", "Water safety", "Contaminants", "Environmental pollution", "Wastewater treatment", "Pharmaceutical residues", "Contaminant removal", "Ecotoxicity". To ensure

the relevance of the included studies, we established specific inclusion and exclusion criteria. Studies were included if they focused on the toxicity of pharmaceutical wastewater, its impact on public health, and associated concerns. Studies reporting on other forms of wastewater or unrelated topics were excluded.

Inclusion Criteria:

- Studies focusing on pharmaceutical wastewater toxicity
- Studies examining the impact of pharmaceutical wastewater on public health
- Studies reporting on the detection and analysis of pharmaceutical compounds in wastewater
- Studies discussing the potential harmful effects of pharmaceutical wastewater contaminants
- Studies investigating the modifying or therapeutic methods for pharmaceutical wastewater

Exclusion Criteria:

- Studies unrelated to pharmaceutical wastewater toxicity
- Studies focusing solely on other forms of wastewater (e.g. industrial wastewater, municipal wastewater)
- Studies not addressing the impact of pharmaceutical wastewater on public health
- Studies with insufficient data or lack of scientific rigor
- Studies just suggest or analysis method(s) for detecting pharmaceutical residues

The relevant information from the chosen studies was extracted and organized based on the key themes and findings related to the toxicity of pharmaceutical wastewater and its implications for public health. This information included details on the types of pharmaceutical compounds that were detected, their potential harmful effects, and any actions taken to address the issue. The extracted data were analyzed and synthesized to provide a comprehensive overview of the current knowledge on the toxicity of pharmaceutical wastewater and its impact on public health. The key findings, trends, and gaps in the literature were identified and discussed in detail.

3. Sources of pharmaceutical residues in wastewater

To implement an effective plan for reducing the concentration of pharmaceutical residue in water, it is essential to have reliable data on the sources that contribute to it (Haguenoer, 2010). These sources can be either natural or human-made (Leckie, 2019). A

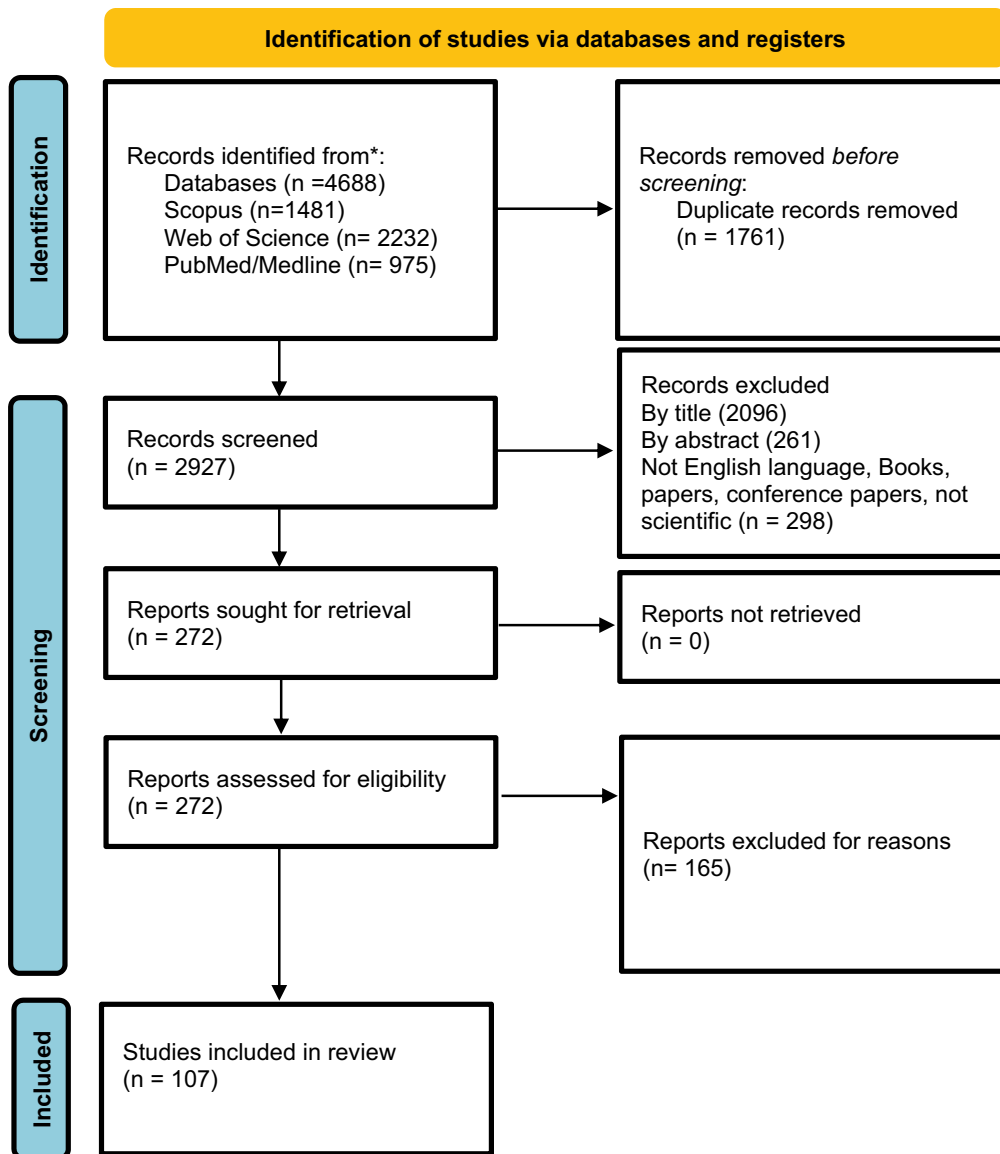


Figure 1. PRISMA Flow chart of article selection.

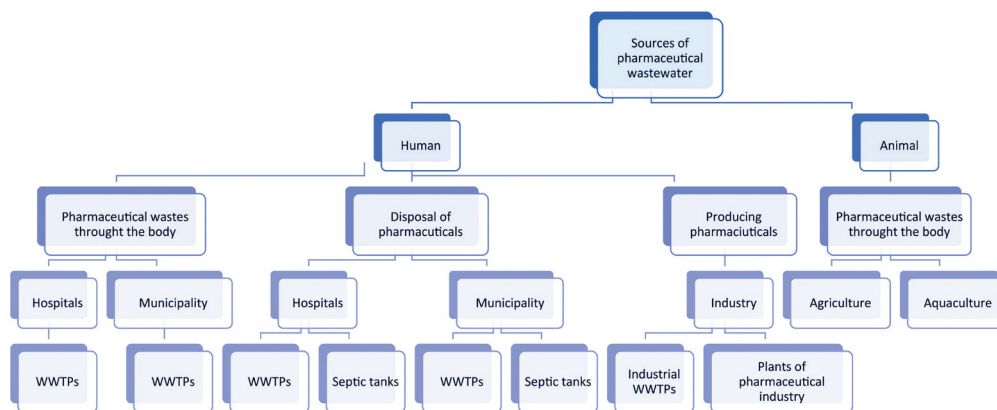


Figure 2. Sources of pharmaceutical residue in wastewater. WWTP = wastewater treatment plants

classification system has been introduced to categorize these sources, as shown in Figure 2. This classification system identifies two main categories of pharmaceutical residue sources in wastewater: a) hospitals and the secondary wastes obtained from therapy and recycling plants; and b) clinical settings in hospitals and domestic areas (Pal, 2018).

According to the Organization for Economic Cooperation and Development (OECD), pharmaceutical residue comes from several sources: 1) pharmaceutical industries, 2) wastewater treatment plants (WWTPs)—which are unable to completely remove waste and are classified into three types: municipal, hospitals', and industries' WWTPs, 3) agriculture and husbandry, 4) aquaculture, 5) septic tanks, and 6) waste management services. Moreover, OECD classifies released pathways into two categories: point sources such as WWTP discharge, and diffuse sources like agricultural runoff (Leckie, 2019).

It is possible to find various pharmaceuticals in catchment surface water, according to the third classification. Sources of emerging organic pollutants (EOSs) include hospitals, livestock farms (due to the excretion of drugs prescribed for animals), fish farms, houses, industrial drug wastes (powdered drugs), and sewage treatment plants (Pal et al., 2010).

Human and veteran body secretions are also known primary sources that release pharmaceutical compounds into watersheds. It is estimated that in humans, 30–90% of orally administered pharmaceuticals are excreted in active forms, mainly through urine and feces (Leckie, 2019; Pal, 2018). Interestingly, the metabolism secretion rate of some drugs, such as gabapentin, is 100% (Madikizela et al., 2017). Some reports showed that pharmaceutical manufacturing facilities and municipal wastewater treatment plants are the primary sources of pharmaceutical contamination (Sági et al., 2022). Pharmaceutical residue found in wastewater has raised concerns, with some reports pointing to agricultural practices and others implicating hospitals and pharmaceutical industries (Ortúzar et al., 2022). Identification of the exact sources of pharmaceutical wastewater would enable such specific approaches as would restrict their impact on the environment and public health and by default interventions that would be both effective as well as efficient in dealing with this critical issue.

4. Typical pharmaceutical compounds found in the wastewater

It is crucial to analyze and address the route of contaminants entry into watersheds, cumulative concentration, and possible toxic effects. The majority of the existing reports found that throughout the world,

hormonal compounds, analgesics, anti-inflammatory products, anti-cancer agents, antibiotics, lipid regulators, beta-blockers, by-products, and even illegal drugs are commonly present at remarkable concentrations in wastewaters. Since most of such chemicals meet one or more criteria set by the National Institute for Occupational Safety and Health (NIOSH) for a hazardous drug, safe handling and disposal of these agents must be carefully monitored (Pal et al., 2010).

4.1. Hormonal compounds

Hormonal compounds are the most pharmaceutical wastes, containing 30% of the reviewed data. They extensively exist in surface and ground raw waters, treated surplus, and agricultural, and rural wastewater treatment facilities (Gabet-Giraud et al., 2010). While these compounds are used for the treatment of endocrine systems disorders in both humans and animals, their residues, even at low concentrations, can impart detrimental effects on the endocrine systems, leading to growing concerns about them in water (Martínez-Alcalá et al., 2018; Thakur et al., 2019; Zheng et al., 2008). Indeed, chemical stability, high lipophilicity, and low water solubility are the characteristics of hormonal substances that result in their accumulation and adverse effects (Shore & Shemesh, 2003). In this regard, the major sources of hormonal wastes are municipal effluents, livestock feeding residues, and secretions (Martínez-Alcalá et al., 2018; Thakur et al., 2019). With respect to the body secretions, it is shown that women's daily excretions contain 10 to 100 µg estrogen, 10.5 µg estrone, and 6.6 µg 17-estradiol (Tiwari et al., 2017).

Estrogen is the most commonly found compound originating from animal manures and humans' excretions (Shore & Shemesh, 2003). Estrogenic compounds or antagonists such as 17α-Ethinylestradiol (EE2) and its metabolite, 17-β estradiol, and Tamoxifen are progesterone compounds found in large amounts in water bodies. Likewise, androgenic compounds such as Testosterone, Beclomethasone, and Hydrocortisone, as well as Zeranone, Trenbolone acetate, and Melengestrol acetate as veterinary growth hormones, are other hormonal substances detected in water sources. Glucocorticoids, phytoestrogen agents (e.g. sesquiterpenes and phytosterols), and other steroidal substances such as synthetic diethylstilbestrol and estriol have also been continuously detected at remarkable concentrations in wastewater (Martínez-Alcalá et al., 2018; Thakur et al., 2019; (Emerging pollutant treatments in wastewater: Cases of antibiotics and hormones; Sacdal et al., 2020).

Several studies reported notable toxic effects produced by hormonal compounds (Emerging pollutant treatments in wastewater: Cases of antibiotics and hormones).

In general, hormonal products can negatively influence the human' reproductive system by affecting the menstrual cycle and causing phenomenal sexual actions and affections. Hormonal wastes also cause the development of tumoral cells in the breast, testis, and prostate and induce cognitive abnormalities in humans (Hassani et al., 2016; Singh & Thakur, 2020). Specifically, estrone has been associated with breast cancer and decreased sperm counts ([Emerging pollutant treatments in wastewater: Cases of antibiotics and hormones](#)). Estrone and EE2 were indicated to have reproductive and developmental abnormalities in various organisms (Adejumoke et al., 2018; Stuart & Lapworth, 2013). Gynecomastia, increases in the rate of breast and testicular cancer, male infertility, and toxic immunologic reactions have been reported as potential risks associated with exposure to estrogen via watersheds (Mohan et al., 2021; Sacdal et al., 2020).

4.2. Analgesics and anti-inflammatory products

Analgesics and anti-inflammatory compounds are applied to eliminate pain and inflammation (Tiwari et al., 2017). As they are available over-the-counter (OTC) in many countries, they must be aware of their possible release into the water. However, our knowledge in this regard is limited. Nevertheless, the most significant sources of opioid analgesics are human body excretions since analgesics are extensively metabolized in humans (Krizman-Matasic et al., 2018; Nielsen & Van Hout, 2015). In this area, the adverse effects of Diclofenac in wastewater on the endocrine system and tissue damage following co-exposure to Acetaminophen, Carbamazepine, Gemfibrozil, and Venlafaxine at 0.5 to 10 µg/L concentrations were shown (Oluwole et al., 2020). Reports have also indicated that metabolites of ibuprofen are more dangerous than their unchanged mother compound (Chopra & Kumar, 2020; Tiwari et al., 2017). The clinical toxicity of each of these agents is comprehensively discussed in Table 1.

4.3. Anti-cancer agents

Chemotherapeutics impart toxic effects on cells' life cycle to avoid their uncontrolled proliferation. Consequently, anti-cancer compounds have carcinogenic, mutagenic, and reprotoxic impact, and thus their wastes may cause similar serious adverse effects on healthy cells of living creatures (Mohan et al., 2021). Information on the actual impact of anti-cancer wastes in water bodies is still limited, but it is confirmed that higher levels of these substances with low degradation

rates produce more hazardous effects. For instance, Vinblastine and Cyclophosphamide are rarely degraded thus they persist and may have considerable dangerous consequences. Moreover, Chlorambucil, Melphalan, and Methotrexate have high concentrations in the aquatic ecosystem due to their high dissociation rates (Singh & Thakur, 2020). Another determining point is the low absorption level, resulting in a lower removal rate. Hence, agents with low adsorption, such as Cyclophosphamide, Ifosfamide, Fluorouracil, and Capecitabine, remain in the aquatic environment for long periods. The primary sources of anti-cancer residues in water bodies are urinary/fecal excretions (Lenz et al., 2007).

4.4. Antibiotics

Natural, semi-synthetic, and synthetic antibiotics are widely used to treat and prevent infections, and they also have some uncommon uses, including growth induction in animals ([Chemical contaminants in feedlot wastes: concentrations, effects and attenuation](#)). Over 30 types of antibiotics are detected in watersheds as the consumption of antibiotics is experiencing a significant increase, primarily due to their over-prescription and the lack of sufficient measures to minimize their use ([Chemical contaminants in feedlot wastes: concentrations, effects and attenuation](#)). Aminoglycosides, Cephalosporins, Penicillins, Tetracyclines, Sulfonamides, and even Beta-lactamase inhibitors are identified in remarkable concentrations worldwide ([Emerging pollutant treatments in wastewater: Cases of antibiotics and hormones](#)); Akash et al., 2020; Hernando et al., 2006; Oluwole et al., 2020).

An augmented concern encompasses antibiotics in water bodies and their possible adverse effects on the environment and human health (SJ Khan et al., 2008). Some researchers found antibiotics as the most abundant pharmaceutical waste in watersheds (Hernando et al., 2006). By 2014, antibiotics were China's most common water pollutants (Cheng et al., 2020).

Regularly, antibiotics are excreted unchanged in the feces and urine of humans and animals. Since more than 80% of antibiotic waste in the USA was used for livestock by 2016, secretions of vertebrae bodies and expired medications are the primary source of antibiotic residue in watersheds (Cheng et al., 2020; SJ Khan et al., 2008).

Hence, there are various adverse effects caused by antibiotics; the potential hazardous effects of antibiotics depend on their molecular structure and physio-chemical features. A worrying increase in antibiotic consumption reflects a severe increment of antibiotic resistance and disruption of biological balance (SJ Khan et al., 2008). According to the US Center for Disease Control and Prevention (USCDC) and the European Center for

Table 1. The most important side effects of commonly found pharmaceuticals in wastewater

Compounds Ref	Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Hormonal compounds		
Reproductive systems' hormonal compounds	17 α -Ethinyl estradiol (EE2)	Advanced bleedings, abnormalities in the reproductive system such as menstrual dysfunctions, uterine, vaginal, and cervical disorders (including bleedings, cramps, and infections), increased sex hormone-binding globulin, libido changes, breast morphological changes, hypertension, pulmonary embolism, galactorrhea, and anxiety (Colli-Dula et al., 2014; Risk et al., 2023; Drospirenone And Ethinyl Estradiol Oral Route)
	17 β -estradiol (E2)	– 4.18–11.76 ng/L, Tehran, Iran, via GC-MS technique, 2014 (Mohagheghian et al., 2014) – 9 ng/L, Canada, 1997 (Huang & Sedlak, 2001) – 0.05–5 ng/ μ L, Taipei, Taiwan, via LC-MS/MS and isotopic dilution technique, 2007 (CHEN et al., 2007) – 0.05–0.07 ng/L, California, USA, via ELISA and GC-MS/MS techniques, municipal wastewaters influent, 2000 (Huang & Sedlak, 2001)
		– 1.02–8 ng/L, Tehran, Iran, via GC-MS technique, 2014 (Mohagheghian et al., 2014) – 9 ng/L, Canada, 1997 (Huang & Sedlak, 2001) – 0.05–10 ng/ μ L, Taipei, Taiwan, via LC-MS/MS and isotopic dilution technique, 2007 (CHEN et al., 2007) – 5.4–7.7 ng/L, Ahvaz drinking water, Iran, 2016 (Hassani et al., 2016) – 13.66 ng/L, Karun River, Iran, via ELISA method, 2016 (Hassani et al., 2016) – 52.71 ng/L, Jiulongjiang River, China, 2014 (Yuan et al., 2014) – 1.4–33.9 ng/L, Dan-Shui River, Taiwan, 2014 – 0.3–7.2 ng/L, Dutch surface water, The Netherlands, 2014 – 0.96 ng/L, Redwood River, The United States of America, 2011 – 4.4 ng/L, Jalle d'Eysines river, France, 2005 (Shore & Shemesh, 2003) – 0.02–0.199 μ g/L, Kwazulu-Natal, WWTPs, South Africa, 2014 – 0.011–0.054 μ g/L Tianjin China, municipal wastewaters influent, 2018 (Wang et al., 2018)
	Tamoxifen	– 740 ng/L, United Kingdom, hospital and urban effluents, 2006 (Z et al., 2013) – 27–212 ng/L, United Kingdom, surface water, 2006 (Orias et al., 2015)
	Testosterone	– 0.119–0.635 μ g/L, Kwazulu-Natal, WWTPs, South Africa, 2014 – 0.026 μ g/L, Kwazulu-Natal, WWTPs, South Africa, 2014 (Manickum & John, 2014)
	Megestrol acetate	– 0.04–0.33 ng/L, River Wenyu, China, 2012 (Chang et al., 2009)
	Beclomethasone	NA
Glucocorticoids (systemic)		– Cortisol and Cortisone: 48.2–130 ng/L, China, 2011 (S Liu et al., 2012) – Triamcinolone: 0.05–12 ng/L, cortisol: 0.6–1.3 ng/L, and Cortisone: 0.08–1 ng/L, Detected from Mühlenbach, River Nahe, Schwelme, River Wupper, Teltow canal, Landgraben, River Neckar, River Lahn, River Rhine, River Ahr, Germany, via HPLC technique, 2018 (Weizel et al., 2018)

(Continued)

Table 1. (Continued).

Compounds Ref	Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Analgesic and anti-inflammatory products		
Acetaminophen	Acetaminophen	– Destruction of electrolyte and molecular balance, oliguria, hepatic malignancies, and anxiety (Ohashi & Kohno, 2020; Gerriets V Nappe TM. AJ. Acetaminophen. [Internet]. Stat Pearls Publishing; Bindu et al., 2020) – Maximum concentration: 1.89 µg/L California groundwater, USA, via HPLC technique, 2004–2011 (Fram & Belitz, 2011) – 0.5–29 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – 1571–37458 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
NSAIDs	Naproxen	– Maximum concentration: 6.84 µg/L, Mbokodweni River, via HPLC technique, South Africa, 2016 (Madikizela & Chimuka, 2017) – 2.6 µg/L, WWTPs, Switzerland, 1999 (Tixier et al., 2003) – 13 ng/dm³, River Cybina water, Poland, via DLLME and LC technique, 2010 – 35 ng/dm³, River Warta water, Poland, via DLLME and LC technique, 2010 (Zgoła-Grześkowiak, 2010) – maximum concentration: 44 ng/L, drinking water, USA, 2008 (Bull et al., 2011) – 1196–5228 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
	Ibuprofen	– Maximum concentration: 19.2 µg/L, Mbokodweni River, via HPLC technique, South Africa, 2016 (Madikizela & Chimuka, 2017) – 1.5–151 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – 1.3 µg/L, WWTPs, Switzerland, 1999 (Tixier et al., 2003) – maximum concentration: 1350 ng/L, drinking water, USA, 2008 (Bull et al., 2011) – 29 ng/dm³, River Warta water, Poland, via DLLME and LC technique, 2010 (Zgoła-Grześkowiak, 2010) – maximum concentration: 4113 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
	Diclofenac	– Maximum concentration: 9.69 µg/L, Mbokodweni River, via HPLC technique, South Africa, 2016 (Madikizela & Chimuka, 2017) – c µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – 0.99 µg/L, WWTPs, Switzerland, 1999 (Tixier et al., 2003) – 5 ng/dm³, River Cybina water, Poland, via DLLME and LC technique, 2010 (Zgoła-Grześkowiak, 2010) – 33 ng/dm³, River Warta water, Poland, via DLLME and LC technique, 2010 (Zgoła-Grześkowiak, 2010) – maximum concentration: 561 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
	Indomethacin	– maximum concentration: 113 ng/L wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
	Acetylsalicylic acid and salicylic acid	– Postoperative bleeding, presyncope, syncope, decreased appetite, heartburn, and increased liver enzymes (Bindu et al., 2020; Lucas, 2016; Munjal, 2023) – Cardiac arrhythmia, disruption in electrolyte balance, metabolic acidosis, gastrointestinal bleeding, perforation, and ulcers, post-term pregnancy, prolonged labor, stillborn infant, low birth weight, brain edema, coma, deafness, acute kidney injuries, and respiratory alkalosis (Bindu et al., 2020, Gąsowska-Bajger et al., 2023; Suwalsky et al., 2013) – 0.05–1.51 ng/L, WWTPs, Germany, 2002 (Szymonik et al., 2017) – 9.5–295 ng/L, Yellow, Hai, and Liao River, China, via HPLC technique, 2008 (L Wang et al., 2010)
	Ketorolac	– 0.2 – 59.5 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – maximum concentration: 2793 ng/L wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
	Ketoprofen	– 0.18 µg/L, WWTPs, Switzerland, 1999 (Tixier et al., 2003) – 19 ng/dm³, River Warta water, Poland, via LC technique, 2010 (Zgoła-Grześkowiak, 2010) – maximum concentration: 801 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)

(Continued)

Table 1. (Continued).

Compounds Ref		Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Carbamates	Meprobamate	Palpitations, syncope, cardiac arrhythmia, tachycardia, ataxia, chills, overstimulation, slurred speech, and severe hypotension (Diseases K. Meprobamate, 2020)	– 43 ng/L, drinking water, The United States, 2008 (Snyder, 2008)
Opioids	Codeine	Cardiac arrest, syncope, ataxia, chills, hallucination, nervousness, blurred vision, and respiratory arrest (Tay & Roberts, 2018)	Maximum concentration: 0.214 µg/L, groundwater, California, USA, via HPLC technique, 2004–2011 (Fram & Belitz, 2011) – 0.01–5.7 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – maximum concentration: 30 ng/L, drinking water, USA, 2008 (Bull et al., 2011)
Anticancer agents			
Chlorambucil		Urticaria, amenorrhea, azoospermia, infertility, anemia, bone marrow suppression and failure, hepatotoxicity, agitation, ataxia, flaccid paralysis, seizure, hallucination, muscle twitching, myoclonus, SIADH (syndrome of inappropriate antidiuretic hormone secretion), toxic epidermal necrolysis, and tremor (Shafqat & Olszewski, 2014, Chlorambucil, 2017)	NA
Melphalan		Decreased appetite, mucositis, dysgeusia, stomatitis, dyspepsia, anemia, decreased white blood cell count, amenorrhea, and renal failure (Ge et al., 2022, Gordillo et al., 2018)	Maximum concentration: 11 ng/L, Catalonia, Spain, via LC-MS/MS and LC-HRMS technique, 2012 (Gómez-Canela et al., 2013)
Methotrexate		Increased liver enzymes, alopecia, burning sensation of the skin, skin photosensitivity, arterial thrombosis, cerebral thrombosis, chest pain, deep vein thrombosis, pericardial effusion, menstrual disease, gingivitis, defective oogenesis and spermatogenesis, urinary tract dysfunctions, vaginal discharge, bone marrow depression, chills, cognitive dysfunction, blurred vision, respiratory failure, severe anemia, exacerbation of hepatitis B, and acute kidney injury (Wang & Peng, 2020, Gaies & Jebabli, 2012)	– 20 ng/L, wastewater, Catalonia, Spain, via SPE-LC-MS/MS technique, 2012 (Negreira et al., 2013) – 0.4–41.3 µg/mL, China, wastewater, via HPLC technique, 2018 (Y Zhao et al., 2019) – 13–60 ng/L, WWTPs, Montréal, Canada, via SPE-LC-MS/MS technique, 2013 (Rabii et al., 2014)
Illegal drugs			
Opioids	Morphine	Urinary retention, atrial fibrillation, bradycardia, facial flushing, syncope, depression, anxiety, abnormality in thinking and dreaming, seizure, agitation, ataxia, chills, decreased cough reflex, malaise, myoclonus, slurred speech, voice disorder, decubitus ulcer, pallor, amenorrhea, gynecomastia, SIADH, delayed gastric emptying, impotence, foot-drop, ostealgia, amblyopia, blurred vision, respiratory insufficiency, orthostatic hypotension, coma, antidiuretic effect, weight loss, toxic megacolon, ejaculatory, decreased bone mineral density, vesicle sphincter, visual disturbance, and apnea (Glare et al., 2006, López-Cano et al., 2023)	– Maximum concentration: 0.3 ng/L, Croatia river, Balkan, 2017 – 10 ng/dm ³ , Rhine River, Germany – 9.8 ng/dm ³ , Ebro River, Spain, 2010 – 16 ng/dm ³ , Henares River, Spain, – 1.4 ng/dm ³ , Barcelona, Spain, 2008 (Szymonik et al., 2017)
Antibiotics			
Aminoglycosides		Reversible nephrotoxicity, vestibular or cochlear ototoxicity (transient or permanent), disequilibrium, lightheadedness, ataxia, and neuromuscular blockade (Krause et al., 2016)	– Streptomycin: 0.8–2.7 ng/ml, – Amikacin: 1–2.3 ng/ml, – Neomycin: 0.4–16.4 ng/ml, – Streptomycin: 0.8–2.7 ng/ml, – Sisomicin: 1–6.7 ng/ml, – Gentamycin: 0.2–2.6 ng/ml, – Paromycin: 0.9–4.2 ng/ml, – Kanamycin: 0.5–7.5 ng/ml, wastewater, via UPLC-MS/MS, Tunisia, 2013 (Tahrani et al., 2016)

(Continued)

Table 1. (Continued).

Compounds Ref	Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Cephalosporins	Agranulocytosis, anaphylaxis, angioedema, arthralgia, cholestasis, genital candidiasis, hepatic failure, increased serum transaminases, exacerbation of myasthenia gravis, interstitial nephritis, decreased hematocrit, decreased hemoglobin, increased lactate dehydrogenase, and increased gamma-glutamyl transferase (DA Khan et al., 2019; Dancer, 2001)	– Cefuroxime: maximum concentration: 141.55 ± 42.48 µg/L – Cefradine: maximum concentration: 20.1 ± 4.02 µg/L – Cefalexin: maximum concentration: 18.48 ± 4.49 µg/L – Cefotaxime: maximum concentration: 18.08 ± 4.02 µg/L – Ceftriaxone: maximum concentration: 15.15 ± 3.17 µg/L – Cefoxitin: maximum concentration: 13.15 ± 3.17 µg/L – Cefazoline: maximum concentration: 12.85 ± 1.56 µg/L, China, wastewater, via HPLC technique, 2016 (Diseases K. Meprobamate, 2020) – Cephalexin: 4.6 µg/L – Ciprofloxacin: 3.8 µg/L – Cefaclor: 0.5 µg/L - Australia, wastewater, via LC-MS/MS technique, 2006 (Watkinson et al., 2007) – Ciprofloxacin: 160–13625 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
Penicillin	Melanoglossia, convulsions, positive direct COOMBS test, serum-sickness-like reaction, vulvovaginal infection, lethargy, myoclonus, seizure, urticaria, flatulence, hairy tongue, increased serum alkaline phosphatase, laryngospasm, serum sickness-like reaction, myoclonus, abnormal platelet aggregation (high doses), and Jarisch-Herxheimer reaction (Bhattacharya, 2010; Fellner, 1986)	Amoxicillin: 0.008–0.035 µg/L Tianjin China, municipal wastewaters influent, 2018 (Wang et al., 2018)
Sulfonamides	Severe allergic reactions, agranulocytosis, Stevens-Johnson syndrome, and hepatic necrosis (Ovung & Bhattacharyya, 2021)	– Sulfadimidine: maximum concentration: 0.005–1.31 µg/L Tianjin China, municipal wastewaters influent, 2018 – Sulfamethoxazole: maximum concentration: 0.012–0.092 µg/L Tianjin China, municipal wastewaters influent, 2018 (Wang et al., 2018) – Sulfamethoxazole: maximum concentration: 0.17 µg/L, groundwater, California, USA, via HPLC technique, 2004–2011 (Fram & Belitz, 2011) – Sulfamethoxazole: 2.5–20 ng/L, drinking water, USA, 2008 (Bull et al., 2011) – Sulfamethoxazole: 162–530 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
Trimethoprim	Maculopapular rash, phototoxicity, pruritus, hyperkalemia, hyponatremia, megaloblastic anemia, methemoglobinemia, neutropenia, thrombocytopenia, increased liver enzymes, increased blood urea nitrogen, increased serum creatinine, and anaphylaxis (Jmw & Juurlink, 2011; Masters et al., 2003)	– 0.385–1.218 µg/dm³, Taff River, UK, 2007 (Szymonik et al., 2017) – maximum concentration: 0.023 µg/L Tianjin China, municipal wastewaters influent, 2018 (Wang et al., 2018) – maximum concentration: 0.018 µg/L, groundwater, California, USA, via HPLC technique, 2004–2011 (Fram & Belitz, 2011) – 0.01–0.03 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – maximum concentration: 1.3 ng/L, drinking water, USA, 2008 (Bull et al., 2011) – 78–197 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
Lincosamide	Abnormal hepatic function tests, abscess at injection site, acute generalized exanthematous pustulosis, agranulocytosis, anaphylactic reaction, anaphylaxis, angioedema, aplastic anemia, azotemia, bullous dermatitis, glossitis, increased serum transaminases, and Stevens-Johnson syndrome (Lincosamides, 2016; Schwarz et al., 2016)	– Lincomycin: 57.8–96 ng/L – Clindamycin: 23.8–26.6 ng/L, Singapore, wastewater, via UHPLC-MS/MS technique, 2016 (Tran et al., 2016) – Clindamycin: –0.039 ± 0.023 µg/L, WWTPs, Brazil, via UHPLC-MS/MS technique, 2017 (Bisognin et al., 2021)
Metronidazole	Vaginitis, genital pruritus, xerostomia, urinary tract infection, pharyngitis, facial edema, flattened T-wave on electrocardiography (ECG), flushing, palpitations, peripheral edema, syncope, and tachycardia (Hernández Ceruelos et al., 2019)	– 1.8–2.9 µg/L, hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) – 44–165 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010) – 0.023 µg/L, WWTPs, Brazil, via UHPLC-MS/MS technique, 2017 (Bisognin et al., 2021) – 8.04 ± 0.56 µg/L, wastewater, Nigeria, via HPLC technique, 2019 (Lan et al., 2019)

Beta-blockers

(Continued)

Table 1. (Continued).

Compounds Ref	Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Beta-blockers (mainly, Propranolol, Betaxolol, Bisoprolol, Atenolol, Metoprolol, Acebutolol, Celiprolol, Metoprolol, Sotalol, Carazolol, Timolol, Nadolol, and Pindolol)	Chronic heart failure reduced left ventricular systolic function, increased peripheral vascular resistance, bradycardia, adverse chronotropic effects, beta-blocker withdrawal, increased sympathetic activity, increased airway resistance, exacerbation of peripheral artery disease, facilitation of hypoglycemia, hyperkalemia, sexual dysfunction (Barron et al., 2013; Ko et al., 2004)	-Atenolol: 0.012–0.409 µg/L - Metoprolol: 0.437–3.21 µg/L Tianjin China, municipal wastewaters influent, 2018 (Wang et al., 2018) - Atenolol: 0.1–122 µg/L - Propranolol: 0.2–6.5 µg/L hospital wastewater, via LC-MS/MS technique, Spain, 2006 (Gómez et al., 2006) - Atenolol: maximum concentration: 26 ng/L, drinking water, USA, 2008 (Bull et al., 2011) -Atenolol: 660–2432 ng/L, -Metoprolol: maximum concentration: 52 ng/L, -Propanolol: 12–61 ng/L, wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010) -Atenolol: Maximum concentration: 46 ng/L, Tone River, Japan, via LC-MS/MS, 2006 (Nakada et al., 2007)
Lipid regulators		
Fibrates	Bezafibrate	Migraine, urticaria, gastritis, flatulence, dyspepsia, increased liver enzymes in serum, increased creatine phosphokinase, bronchitis, pharyngitis, acute renal failure, alopecia, cholelithiasis, decreased gamma-glutamyl, transferase, reduced platelet count, decreased serum alkaline phosphatase, decreased white blood cell count, erectile dysfunction, rhabdomyolysis, skin photosensitivity, Stevens-Johnson syndrome, and toxic epidermal necrolysis (Goldenberg, 2008; Ellis et al., 1994)
	Gemfibrozil	Atrial fibrillation, angioedema, arthralgia, blurred vision, bone marrow, depression, decreased libido, hypokalemia, impotence, increased creatine phosphokinase, increased serum alkaline phosphatase, increased serum bilirubin, increased serum transaminases, laryngeal edema, myopathy, and Raynaud phenomenon (Ahmed et al., 2021; K Kim et al., 2017)
Statins	Mainly Atorvastatin, Fluvastatin, Simvastatin, Lovastatin, Pravastatin	Hepatic dysfunction, muscle injury, hypothyroidism, renal dysfunction, behavioral and cognitive, diabetes mellitus, increased risk of cancer, cataract, neuropathy, lupus, androgen synthesis, and immune response (Ruscica et al., 2023)
	Ezetimibe	Increased serum transaminases (with HMG-CoA reductase inhibitor), arthralgia, anaphylaxis, angioedema, autoimmune hepatitis, increased creatine phosphokinase, myalgia, myopathy, rhabdomyolysis, thrombocytopenia, and urticaria (Wang et al., 2022; Florentin et al., 2007)
Others		
Furosemide		Orthostatic hypotension, bullous pemphigoid, bladder spasm, blurred vision, destruction of ionic compounds balance in blood, deafness, and hepatic encephalopathy (X Dai et al., 2021; Wola et al., 2022)

(Continued)

Table 1. (Continued).

Compounds Ref	Side effects (The mentioned side effects are the most possible and noticeable ones)	Examples of reported concentration
Hydrochlorothiazide	Alopecia, dermatologic abnormalities, aplastic and hemolytic anemia, blurred vision, destruction of ionic compounds balance in the blood, sialadenitis, and angle-closure glaucoma (Thiazide diuretics, 2016; Herman & Weber, 2023; Opsha, 2015)	$-1.7 \pm 0.2 \mu\text{g/L}$, Kuopio, Finland, WWTPs, via HPLC-MS/MS technique, 2019 (Ikonen et al., 2021) $-617\text{--}10,018\text{--}1051\text{ng/L}$ wastewater, Spain, via GC-MS technique, 2008 (Rosal et al., 2010)
Disopyramide	Xerostomia, urinary hesitancy, cardiac conduction disturbance, syncope, malaise, myasthenia, nervousness, flatulence, impotence, urinary frequency, retention, urgency, and blurred vision (Jh & An, 2016; Disopyramide, 2006)	Maximum concentration: 19 ng/L, Tone River, Japan, via LC-MS/MS, 2006 (Nakada et al., 2007)

Abbreviations: NA: Not available; NSAIDs: Nonsteroidal anti-inflammatory drugs; UK: United Kingdom; USA: United States of America, LC-MS/MS: Liquid chromatography-mass spectrometry, GC-MS: Gas chromatography-mass spectrometry, WWTPs: Wastewater treatment plants, HPLC-MS/MS: High-performance liquid chromatography-mass spectrometry, HPLC: High-performance liquid chromatography, UHPLC-QTOF-MS: Ultra-high-performance liquid chromatography-triple/time-of-flight mass spectrometry, ELISA: Enzyme-linked immunosorbent assay, SPE-LC-MS/MS: solid phase extraction coupled with liquid-chromatography tandem mass spectrometry.

Disease Prevention and Control (ECDC), antibiotic resistance is responsible for more than 23,000 and 25,000 deaths per year in the USA and Europe, respectively. Importantly, it was indicated that this phenomenon is responsible for more than 750,000 deaths per year worldwide ([Antibiotic Resistance: A Global Threat \[Internet\]; Infographic: Antibiotic Resistance The Global Threat](#)). Besides microbial resistance, chloramphenicol may lead to chromosomal abnormalities (Munita et al., 2016).

4.5. Lipid regulators

Bezafibrate, Clofibric acid, Gemfibrozil, Atorvastatin, Fluvastatin, Simvastatin, Lovastatin, Pravastatin, Fenofibric acid, and Ezetimibe are commonly found in water bodies at remarkable concentrations (Hernando et al., 2007, 2006; Nassiri Koopaei & Abdollahi, 2017; Szymonik et al., 2017). Lipid regulators are one of the most prescribed drugs worldwide that control the lipid profile in human and animal bodies (Zhang et al., 2020).

Several toxicological effects of lipid regulators were reported, but information on their particular consequences on human health through exposure to water bodies is insufficient (Hernando et al., 2007; Zhang et al., 2020).

4.6. Beta-blockers

Beta-adrenergic antagonists or beta-blockers are commonly used for treating arterial hypertension, anxiety, angina, cardiac dysrhythmia, and coronary artery diseases (Ramil et al., 2010; Szymonik et al., 2017), and their use may vary among different regions, leading to

alternations in their existence in the wastewater. For instance, it was estimated that consumption of beta-blockers in Germany is above 100 tons per year or that over 35 tons of Propranolol was taken in France in 1999. Based on the literature, beta-blockers generally exist in treated wastewater, surface waters, municipal effluents, and hospital wastes (Miège et al., 2006; Ramil et al., 2010). Beta-blockers are detected in municipal wastewater as they are partially metabolized in human bodies (Miège et al., 2006).

Noteworthy, the accumulation of these chemicals in water bodies and food chains produces severe physiological adverse effects against aquatic livestock, animals, and humans (Gabet-Giraud et al., 2010; Ramil et al., 2010; Wick et al., 2009). The most common beta-blockers detected in wastewater are Propranolol, Betaxolol, Bisoprolol, Atenolol, Acebutolol, Celiprolol, Metoprolol, Sotalol, Carazolol, Timolol, Nadolol, and Pindolol (Akash et al., 2020; Hernando et al., 2006; Nassiri Koopaei & Abdollahi, 2017; Oluwole et al., 2020; Wick et al., 2009).

4.7. By-products

Principally, the human and animal bodies metabolize drugs, resulting in biochemical changes in their structure; then, the metabolites or by-product compounds are exerted in urine or feces that finally run into water resources. Studies detected remarkable concentrations of Norfluoxetine and Desmethylsertraline as metabolites of Fluoxetine and Sertraline in watersheds. These metabolites are 50% more toxic than the mother compound. Furthermore, metabolites of Phenazone, Propyphenazone, Acetylsalicylic acid (Salicylic acid and Gentistic acid), Ibuprofen (1-hydroxyl

Ibuprofen and carboxy-ibuprofen), Citalopram (Desmethylcitalopram), Carbamazepine (32 metabolites), and Sulfamethoxazole were detected in high concentrations in surface waters.

Although little information is available, 10,11-dihydro-10,11-epoxy Carbamazepine, N4-Acetylsulfamethoxazole, and 4-hydroxy diclofenac can potentially inhibit aquatic organisms' immune systems by making strong conjugation with cells' proteins (Table 1).

4.8. Illegal drugs

The intensive international use of illegal drugs and their frequent unregulated discharge into water bodies are worldwide issues nowadays. A class of medications that is not used for medical targets and is prohibited mainly by either national or international legislation is called illegal drugs (Mohan et al., 2021). Narcotics are used for either medical or abuse purposes. In this regard, essential medical indications for illicit drugs are pain/cough relief, laryngology, ophthalmology, local anesthesia, and attention deficit hyperactivity disorder (ADHD) treatment (Szymonik et al., 2017). Narcotics that are most commonly found in watersheds are amphetamine, amphetamine-sulfate, opiates, morphine, cannabinoids, and cocaine (Mohan et al., 2021). Moreover, it is estimated that the amount and distribution of 4-(Methylnitrosamino)-1-(3-pyridyl)-1-butanone will increase in the future, particularly in countries with higher rates of smoking (Lorenzo & Picó, 2019).

The most critical side effects reported for commonly discovered compounds and their concentration in watersheds are summarized in Table 1. The mentioned adverse effects might challenge human health due to the expanded use and release of medications into water bodies and, hence, their entrance into human bodies via different sources.

5. Recycling

Sources of fresh water are limited and cannot afford the needs of the world's current population. Thus, recycling water, particularly pharmaceutical wastewater, has been suggested over recent decades (Klemeš, 2012; Musie & Gonfa, 2023). However, due to some limitations, including a shortage of budget for or access to appropriate recycling methods, interregional variations exist in this regard. For instance, it was shown that countries with the highest level of treatment of water bodies are in the following order: North Atlantic, Baltic Sea, and Western Europe countries. Conversely, watersheds of the

Caribbean, Caspian Sea, East Asia, Southern Asia, and West and Central Africa are the least treated (Gadipelly et al., 2014).

Several studies discussed efficient recycling methods to remove pharmaceutical residue even at low levels (Oulebsir et al., 2020) and to make the water useable/potable (Pratyusha et al., 2012). Recent research has raised concerns over the long-term effects of these micropollutants on human health, showing the presence of substances like hormones and pharmaceuticals in water. Conventional water treatment techniques like chlorination and coagulation have not successfully removed these pollutants. However, cutting-edge methods like reverse osmosis, granular activated carbon, and nanofiltration have effectively eliminated over 95% of these chemicals from water (SD Kim et al., 2007). Therefore, several water treatment methods have been introduced. Nonetheless, two essential points must be taken into account: first, the elimination of hydrophobic pharmaceutical agents from watersheds cannot probably prevent the accumulation of toxic agents in biosolids, and second, most pharmaceutical contaminants are not eliminated from watersheds via conventional methods (37, Wick et al., 2009).

Various treatment techniques are used to eliminate pharmaceutical wastes from waterbodies, including physical, chemical, and biological methods (Rodríguez-Serin et al., 2022). Moreover, pretreatment methods to recycle and reprocess byproducts, including solvents, acids, heavy metals, and active pharmaceutical ingredients, are suggested (Renita et al., 2017).

Physical methods such as adsorption, zeolite, and carbon nanotubes are primarily used to remove these contaminants (Mishra et al., 2023). Adsorption with activated carbon or zeolite can efficiently remove trace pollutants, but it can remove pollutants non-selectively, and the applied system needs to be re-administered. Carbon nanotube is the most efficient method of adsorption, but it is expensive and needs to be readministered (De Luca et al., 2021).

Chemical methods like advanced oxidation processes (AOPs) and photocatalytic degradation techniques have also been used to remove these contaminants (Cardoso et al., 2021). Biological methods include biodegradation, which uses microorganisms to degrade pollutants, and phytoremediation, which involves using plants to remove contaminants from water (AU Khan et al., 2022).

In recent studies, some innovative methods have been introduced to be effective for the removal of drug-related pollutants in water consisting of metal-based nanohybrids (metals/metal oxides) and carbon-based nanohybrids (carbon nanotubes, graphene,

fullerenes, etc.; Javaid et al., 2022), metal-organic frameworks (MOFs) materials as adsorbents or photocatalysts for wastewater treatment (Kairigo et al., 2020), zinc ferrite nanoparticles with enhanced photocatalytic performance (Latif et al., 2023) and MXene-based hybrid composites (Latif et al., 2023). In this regard, Table 2. enlists different confirmed methods focusing on their pros and cons.

6. Pharmaceutical wastewater disposal: Need for proper education

There are two crucial steps in clearing the water from pharmaceutical wastes and preventing their hazardous effects: minimizing their release and safely removing them from contaminated water. Studies confirmed the positive impact of proper education on both stages (Paut Kusturica et al., 2020).

Lack of knowledge on severe adverse effects that could be potentially caused by exposure to pharmaceutical residue in water is the main reason behind the global ignorance—as it is not limited to the general population or non-experts—of their poor disposal and removal (Luis et al., 2020; Paut Kusturica et al., 2020). There are some reports that pharmacists and medical staff are not adequately concerned about the potential hazard of pharmaceutical residue in water bodies (Aditya & Rattan, 2014). One of the essential sources of pharmaceutical waste's entrance into water bodies is household effluents. Studies indicate that expired drugs are restored in houses primarily because the treatment protocol was not completed or their household disposal methods are not precise (Paut Kusturica et al., 2020). Most people do not know what to do with expired medications, unlike those who are properly trained to store them (Maharaj et al., 2020; Paut Kusturica et al., 2020). In this sense, there is an urgent need for adequate training on scientific and efficient removal of pharmaceuticals (Maharaj et al., 2020).

It is not confirmed whether pharmacies or the public population are responsible for treatment costs. Yet, studies reported that most people are not interested in paying the expenses. However, proper education regarding the importance of environmental balance and the potential adverse health effects of exposure to such chemicals via water plays a significant role in people's preferences. Unfortunately, pharmacy schools do not have enough efficient courses about properly disposing of pharmaceutical wastes. Some experts believe that healthcare professionals and related stakeholders are responsible for developing measures to increase public information (Maharaj et al., 2020).

Cultural impressions may impact the prescription of medications or disposal of pharmaceutical wastes into watersheds (Thomas, 2017). Generally, pharmacists play vital roles in minimizing the level of pharmaceutical wastes in water bodies, thereby minimizing their toxicity. Thus, pharmacists can help ameliorate drug prescription practice, drug consumption/application, and their safe disposal. Increasing public awareness about hazardous severe effects of pharmaceutical wastewater and teaching proper disposal methods to patients and healthcare workers may also help reduce the residues levels of such chemicals in water bodies (Aditya & Rattan, 2014; Singleton et al., 2014).

7. Role of regulators in the management of pharmaceuticals residue

Better health care available in developing countries and increased medicine consumption have resulted in higher amounts of pharmaceutical residue being released into the environment. Thus, if reasonable measures (i.e. proper regulations) are not taken against this issue, contamination will be progressively continued and worsened (Aus der Beek et al., 2016). Global countries have different levels of legislation and regulation (De Freitas & Radis-Baptista, 2021). Although considerable deficits often exist in regulations set in some regions, particularly developing countries, in others, more advanced legislation is present, especially in Canada, Australia, the USA, and European countries. In contrast, there are no specific regulations in most low- and middle-income nations like South Africa (Maycock & Watts, 2011; Ngqwala & Muchesa, 2020). Different approaches regulated to minimize pharmaceutical wastewater levels are summarized in Figure 3.

At a national level, the management of pharmaceutical residue can be classified into three sections: 1) source-directed approaches, 2) user-oriented approaches, and 3) end-of-pipe. These policies employ regulatory, economic, and voluntary guidelines. Policies related to pharmaceutical wastewater are at the end-of-pipe policies level. These policies regulate the pharmaceutical life cycle's collection and disposal, wastewater treatment, and reuse approaches. End-of-pipe policy centers eliminate drugs and their metabolites after utilization or delivery into the water, like norms on proper waste removal (Pharmaceutical Residues in Freshwater, 2019). Such policies intend to control pharmaceuticals residue levels in the environment by regulating one or multiple stages of the pharmaceutical life cycle that are as follows: 1) design; 2) marketing authorization; 3) production; 4) post-authorization; 5) prescription and consumption; 6) collection and disposal; and 7)

**Table 2.** Comparison of proven treatment methods to eliminate pharmaceutical wastes in waterbodies

Physical methods	Adsorption	Activated carbon	Mechanism and classes	Agents	Pros	Cons
			Processed carbon with small interstices and a large surface area, which increases the rate of adsorption, is used. The effect of activated carbon depends on pharmaceuticals' hydrophobicity and charge features (Renita et al., 2017).	Antibiotics, Estradiol, Estriol, Anti-Diarrheal, Loperamide, Nitroimidazole, Sulfamethoxazole, Tylosin, Tetracycline, lipid regulator, Bezafibrate, Gemfibrozil, Thimerosal, Propranolol, Diclofenac, Paracetamol, Ketoprofen, Fenoprofen, Naproxen, Ibuprofen, Diazepam, Thioridazine, Tamoxifen, Dilantin, Carbamazepine, and Primidone (Pratyusha et al., 2012; Barceló & Petrović, 2008)	-Efficient for trace pollutants. -Requires low budget, low energy, and little time. -Fully attainable and practical (Barceló & Petrović, 2008; Zaid et al., 2020). -Chemically sustainable. -Removes pollutants non-selectively (Guo et al., 2017).	After implementation, the applied system needs to be readministered (Barceló & Petrović, 2008)
		Zeolite	Some aluminosilicate substances are used to catalyze the process of adsorption in zeolite method. Over 40 types of natural and 100 types of synthetic zeolite methods (Barceló & Petrović, 2008).	Acetylsalicylic Acid, Amoxicillin, Atenolol, Diclofenac, Ibuprofen, Inorganic, Naproxen, Organic Agents, Salicylic Acid, and Tetracycline (Barceló & Petrović, 2008)	-Efficient for trace pollutants. -Fully attainable and practical (Barceló & Petrović, 2008) -Requires low energy (Zaid et al., 2020)	-Not as effective as other adsorption methods. -After implementations, the applied system needs to be readministered (Barceló & Petrović, 2008)
		Carbon nanotube	One or several layers of hollow graphene sheets are the basis of this method (Renita et al., 2017).	Dichlorophenol, Amoxicillin, Cephalixin, Diclofenac, Cefazidime, Ofloxacin, Tetracycline, Carbamazepine, Lincomycin, Sulphapyridine, Norfloxacin, Oxytetracycline, Sulfamethoxazole, Ibuprofen, Acetaminophen, Caffeine, Prometryn, Carbendazim, P-Nitrophenol, and Ciprofloxacin (Renita et al., 2017).	-Suitable for trace pollutants. -Due to its high surface area, it is known as the most efficient method of adsorption class. -Fully attainable (Renita et al., 2017). -Requires low energy (Zaid et al., 2020). -An Eco-friendly method.	-High cost. -After implementations, the applied system needs to be readministered (Renita et al., 2017).
		Bio-sorbents	Agricultural originated	Amoxicillin, Cephalixin, Carbamazepine, Ciprofloxacin, Diclofenac, Dorzolamide, Ibuprofen, Ketoprofen, Metronidazole, Naproxen, Norfloxacin, Paracetamol, Penicillin G, Pramipexole, Sulfamethoxazole, Sulfonamides, and Tetracycline (Renita et al., 2017).	-It is low-cost and highly accessible (Renita et al., 2017). -It has an essential operation, composing little number of wastes, and high efficacy. -It can remove pollutants at deficient levels (Kanamalapudi et al., 2018).	NA
			Cork powder and granules originated	Several antibiotics and anti-depressants, Ibuprofen and Paracetamol (Renita et al., 2017).		
			Chitosan originated	Dorzolamide and Pramipexole Dihydrochloride (Renita et al., 2017).		

(Continued)

Table 2. (Continued).

Mechanism and classes			Agents		Pros	Cons
Industrial residues			Macrolides, Quinolone Antibiotics, Ciprofloxacin Hydrochloride, Pentachlorophenol, Sulfonamides, and Trimethoprim (Renita et al., 2017).			
Electrodialysis	The electrodialysis technique is a mixture of electrolytic and dialysis ambiguity (Guo et al., 2017)	Ninety-seven percent of contaminants are ionic compounds and dissolved solids (Moravvej et al., 2020).			-Requires little energy. -Simple administration. -composing a small number of pollutants (Guo et al., 2017). -Efficient for water-softening purposes - Requires little space (Grégorio Crini & Lichtfouse, 2019). Various evaporation treatment methods specifically affect each type of contaminant (Grégorio Crini & Lichtfouse, 2019).	Fouling (Akter & Park, 2023) -May lead to lower pH. -High cost. -Procedures make machines polluted (Zaied et al., 2020) -It is costly for high amounts of untreated water (Grégorio Crini & Lichtfouse, 2019).
Reverse osmosis	Cellulose ester Aromatic polyamide	Reversing the original pathway of water's flow in the osmosis process is called reverse osmosis. Pre-filter, post-filters, and catalysts empower this process (CDC, 2008).		Common chemical contaminants (metal ions, aqueous salts; CDC, 2008)		
Evaporation	Evaporation removes contaminants by remaining liquids with an upper boil point while evaporating the main solute (Snyder, 2008).	NA				
Filtration	removing suspended particles as they pass through a bed of granular media, primarily by physical action (Cescon & Jiang, 2020).	Turbidity, color, microbes, and particulates—whether they are created by pre-treatment or already existing in the water (Cescon & Jiang, 2020).		-Simple operation (Grégorio Crini & Lichtfouse, 2019).		-High cost (Grégorio Crini & Lichtfouse, 2019).
Sedimentation	Gravity power, based on the density of pollutants, poses the sedimentation of substances (Guo et al., 2017).	Settleable solids such as suspended ones (Chatzakis et al., 2006).		-Simple administration and comprehensive technology (Guo et al., 2017).		-Not efficient for organic compounds that are fully dissolved (Guo et al., 2017). - Non-ionizable compounds cannot be removed through this method (Vieno et al., 2006).
Coagulation	The addition of chemical reagents that facilitate flocculation, mixing, and eventually stabilizing contaminated water poses flocculation (Guo et al., 2017).	Bezafibrate, Diclofenac, and Ibuprofen (Vieno et al., 2006).				
AOP	Ozone or hydrogen peroxide	In ozonation, OH active radicals oxidate considered contaminants non-selectively (Renita et al., 2017)	Amoxicillin, Estrogen, and Penicillin (Jaiswar et al., 2022;Lucas, 2016)	-Effectively facilitate sterilization of materials (Zaied et al., 2020). -Remove low biodegradable agents (Gadipelly et al., 2014) -Ozone is a significantly powerful oxidant (Guo et al., 2017)		-High cost and energy demand. Composing toxic mediators such as active radical agents (Renita et al., 2017;Zaied et al., 2020) -Compounds that contain active amide groups are resisted by this method (Gadipelly et al., 2014).
	Photolysis	Fenton oxidation Iron salts and hydrogen peroxide facilitate considered contaminant elimination by breaking down	Cyclohexanone, Carbamazepine, and Hydroxylamine, Penicillin, Pyridine and Toluene (Gadipelly et al., 2014).	-Remove low biodegradable agents. -Due to iron's extensiveness, this method is globally		-High cost and energy demand. -May need pre-treatments to improve the elimination rate (Renita et al., 2017;Zaied et al., 2020)

(Continued)

Table 2. (Continued).

Mechanism and classes		Agents	Pros	Cons
Hydrogen peroxide and composing hydroxyl radicals as catalysts of reactions (Renita et al., 2017).			Available (Gadipelly et al., 2014).	
UV treatment	Direct UV lights decompose the chemical bonds of pollutants (Renita et al., 2017).	Ciprofloxacin, Cyclophosphamide, and Trimethoprim (Pratyusha et al., 2012).	-Mostly applied after either biological treatments or sand filtrations (Phillips et al., 2010). -Removes low biodegradable agents (Gadipelly et al., 2014). -Requires little energy. -Eco-friendly. -Low cost. -Bio-activity (Lenz et al., 2007; Li & Li, 2015).	-Scattering of light. -Spontaneous e^-/H^+ recombination -Photo corrosion phenomenon. -Limited effect on pharmaceutical agents (Majumder et al., 2020).
TiO ₂ and ZnO catalysis	There are various catalysis-based devices, considering each device's specific features (Majumder et al., 2020).	Ampicillin, Carbofuran, Diclofenac, Ibuprofen, Lindane, Metronidazole, Ciprofloxacin, Paracetamol, and Chloramphenicol (Majumder et al., 2020).		
Electrooxidation	The Electrooxidation method is based on OH radical formation due to the high oxidizing power of this agent (Gadipelly et al., 2014)	Carbamazepine, Diclofenac, Rthinyloestradiol, Ibuprofen, and Propranolol (Gadipelly et al., 2014)	-Non-selective removal of compounds (Gadipelly et al., 2014)	
Wet air oxidation	Like other oxidation methods, the wet air technique is based on OH radical formation facilitated by high temperatures. Moist air plays a catalyst role in this process (Gadipelly et al., 2014).	NA	-Fast-acting (Grégoire Crini & Lichtfouse, 2019).	-This method generates different kinds of metabolites (Grégoire Crini & Lichtfouse, 2019).
Ultrasound oxidation	Ultrasonic waves by physicochemical processes decompose water contaminants. Processes are mixed with free radical oxidation, pyrolysis, and supercritical water oxidation techniques (Guo et al., 2017).	Diclofenac, Ibuprofen, and Triclosan (Pratyusha et al., 2012).	-Biocompatible (Pratyusha et al., 2012).	-High cost -Time- consuming (Pratyusha et al., 2012).
Electrochemical oxidation	Through electrochemical reactions, pollutants become oxidized and thus degraded (Guo et al., 2017).	Cyclophosphamide, Fluorouracil, Imatinib, Ifosfamide, and Methotrexate (Siedlecka et al., 2018).	-Eco-friendly. -Highly efficient. -No requirements of either particular temperature or pressure. -Little required space. -No intermediate pollutant formation (Guo et al., 2017).	Electrochemical oxidation can form by-products that may reduce the efficacy of the treating process (Grégoire Crini & Lichtfouse, 2019).

(Continued)

Table 2. (Continued).

Mechanism and classes			Agents	Pros	Cons
Others	Chlorination	In conventional water treatment methods, chlorine removes microorganisms and disinfects the water. However, chlorine can also react with pharmaceuticals to eliminate them from water sources. This elimination can be thoroughly done or end up with safe by-products (Rodil et al., 2012).	17-A-Dihydroequilin, 17A-Ethinylestradiol, 17β-Estradiol, Estriol, Erythromycin, Estradiol, Medrogestone, Norgestrel, Sulfamethoxazole, Trimegestone, Trimethoprim, and Valerate (Organization WH, 2012).	-Fast-acting (Rodil et al., 2012). -Simple application. -Low cost (Nielsen et al., 2022)	-This method may result in different types of metabolites (Rodil et al., 2012). -Changes in taste and odor of the treated water (Grégorio Crini & Lichtfouse, 2019). -High cost -Low efficiency of regeneration -complex operation. Some contaminants cannot be degraded due to the limited generation of reactive hydroxyl radicals (W Dai et al., 2023). -High cost (Pratyusha et al., 2012).
	Pilot-scale integrated process	A novel method for treating pharmaceutical wastewater combines CSTR, MECs, EGSB, and MBBR. The EGSB + MBBR unit, the primary biochemical process at the back end, removes most of the nutrients and reduces certain pollutants (W Dai et al., 2023).	Benzothiazole, Pyridine, Indole, and Quinoline, COD, NH_4^+ -N and TN (W Dai et al., 2023)	High removal rates of toxic pollutants (W Dai et al., 2023)	
	Plasma treatment Thermal Non-thermal	In plasma treatment, a remarkable amount of energy is transferred to pollutants to improve their contact and sedimentation rate (Pratyusha et al., 2012; Samal, 2017).	NA	-Requires little energy (Pratyusha et al., 2012).	

Abbreviations: AOP: Advanced oxidation process; MIP: Molecularly imprinting technology; COD: chemical oxygen demand, NH_4^+ -N: ammonia nitrogen, TN: total nitrogen; NA: Not available

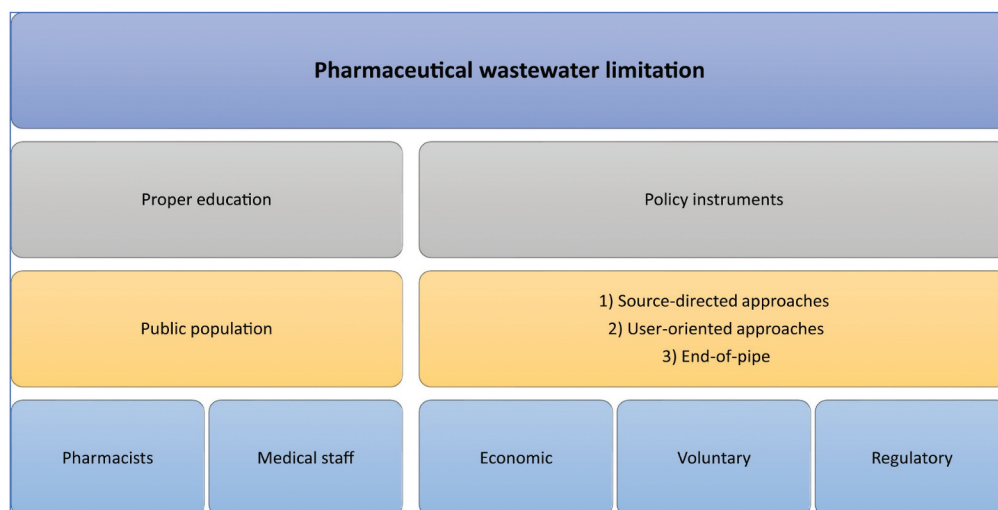


Figure 3. Ways of limiting pharmaceutical wastewater entrance to the environment. Note: Proper education and policymaking are two crucial activities that can reduce pharmaceutical wastewater. Good education, through instructions provided by pharmacists and medical staff, directly affects the general population. Policy instruments are implemented in three sections: 1) source-directed approaches, 2) user-oriented approaches, and 3) end-of-pipe. Three policy instruments for limiting pharmaceutical contaminations include regulatory, economic, and voluntary measures.

wastewater treatment and reuse. Mitigation options can be upgrading design, production, WWTPs, etc. Some countries, like the UK, use different mitigation options. “Strategic Approach to Pharmaceuticals in the Environment” distinguishes activities for partners involved in the pharmaceuticals life cycle with an accentuation on sharing excellent performance, collaborating at a global level, and improving comprehension of the risks at the European Union (EU) level. WWTP improvement and upgrades with new technologies are key mitigation options (Pharmaceutical Residues in Freshwater, 2019).

The occurrence of pharmaceuticals in the environment led to the development of regulations in environmental risk assessment of pharmaceutical residues to address concerns about this problem. The increase of pharmaceutical residue occurrence in the environment was recognized as an emerging environmental problem when the European Commission (EC) invested in a scientific project on this topic; the results of this project increased public awareness, and then the European Environment Agency (EEA) acknowledged this problem (Küster & Adler, 2014). The WHO reports that controlled chemicals and cytotoxic medications are now subject to stricter laws than ordinary medications. However, wastewater disposal is not prohibited. Non-controlled chemical disposal is more varied and frequently done at municipal, jurisdictional, or regional levels (Maycock & Watts, 2011).

The European Union has a specific approach to reducing environmental pharmaceutical residue. The European Union Strategic Approach to Pharmaceuticals in the Environment (PiE) was adopted in March 2019 to deal with the environmental consequences of all stages of the pharmaceuticals lifecycle. Subsequently, the environment council indicated that necessary actions should be taken to decrease pharmaceuticals and their residue risks. In September 2020, a resolution was adopted by the European Parliament to develop an integrative approach to reduce the dangers of pharmaceutical residues. The status of the EU strategic approach to PiE actions is classified into four categories, from which, currently, 3 actions have been implemented, 15 actions are ongoing, 5 actions are in a good progress situation, and 10 actions are started (European commission, 2019). The issue of environmentally persistent pharmaceutical pollutants (EPPPs) was identified as a major concern during the Strategic Approach to International Chemicals Management (SAICM) Fourth International Conference of Chemicals Management (ICCM4) in 2015. Recognizing the gravity of the situation, ICCM4 called for international cooperation to take immediate and effective action. Furthermore, the conference stressed the importance of spreading awareness about EPPPs and enhancing access to related information. ICCM4 recommended policymakers and stakeholders to work in unison to deepen their knowledge of EPPPs by taking collaborative measures (Miettinen & Khan, 2022).

Despite all policies and regulations set in this context, levels of pharmaceutical residue in the environment are still high, indicating that these regulations have probably not been appropriately applied (Aus der Beek et al., 2016).

8. Suggestions and further perspectives

A considerable number of studies determined sources of pharmaceutical residue in the environment and their class, toxic effects, and exact concentration, as well as practical disposal/removal approaches and regulatory methods. Nevertheless, there are still undeniable gaps in the current knowledge about pharmaceutical waste. For instance, there is no confirmed data about the toxic effects of long-term exposure to some of the most commonly found contaminants (e.g. anti-cancer products, lipid regulators, and beta-blockers) via water consumption. Moreover, in low- and middle-income countries, the exact concentration of most pharmaceutical pollutants is unknown. Thereby, conducting further monitoring studies is warranted to provide data for ecotoxicological risk assessments. A better way to address environmental pollution's complex problems may be by combining analytical chemistry and toxicity identification evaluations (Barceló & Petrović, 2008).

The occurrence of pharmaceutical residue in the environment is increasing, and this potentially increases the need for water treatment. The high water treatment costs would make providing clean water more difficult; thus, poor people try to use cheaper/unhygienic water. Legislators in developing countries should prevent routing pharmaceuticals and chemicals to the environment by regulating the disposal of industrial, livestock, municipal, and other sources of wastewater and contaminants (Kamba et al., 2017).

OECD suggests that governments use the life cycle approach to reduce pharmaceuticals in the environment. This approach requires 1) planning and carrying out a blend of source-directed, user-orientated, and end-of-pipe measures policies; 2) focusing on stakeholders that play roles in the life cycle of pharmaceuticals; and 3) utilizing a mix of intentional, financial, and administrative instruments (Pharmaceutical Residues in Freshwater, 2019). It should be noted that there is currently no regulatory framework to control pharmaceutical disposal, and adequate legislation to monitor the appropriate disposal of these chemicals is needed (Jose et al., 2020).

9. Conclusion

Pharmaceutical wastewater is one of the global issues that threatens public health nowadays. To better solve

this problem, it is vital to understand the various kinds of pharmaceutical waste and the hazardous amount of toxicity. In this regard, the more multi-centric studies have been conducted, the better we can confront with this worldwide issue. Nevertheless, suitable legislation about the disposal of pharmaceutical waste is also necessary. The effect of educational support is undeniable, and if we raise public awareness about environmental health through healthcare providers and professionals to participate in preventive strategies, we can achieve successful pharmaceutical waste management. International public health campaigns can also voluntarily focus on green manufacturing practices in the pharmaceutical industry and encourage consumers to use green medication. In addition, more comprehensive studies are needed to understand the long-term health impacts of pharmaceutical contaminants, especially in developed countries.

On the other hand, developing innovative and efficient wastewater treatment methods specifically targeting pharmaceutical contaminants can help us to alleviate the dangerous influence of pharmaceutical wastewater on the environment, as the conventional treatment processes have limitations in removing these compounds effectively. So that, new technologies for the recycling pharmaceutical wastewater methods should be explored and implemented.

In the end, it is fundamental to know that if we want to achieve the most effective solution to decrease the pharmaceutical wastewater toxicity, a multi-disciplinary approach and a practical collaboration among the governments, researchers, healthcare practitioners, and the public is required.

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All authors contributed significantly to the present work regarding the conception, study design and execution, interpretation of the findings, and revising/critically reviewing the article. All authors approved the present version to be published. Authors agree on the journal to which the article has been submitted, and they are accountable for all aspects of the work.

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